

React Native Setup & Login App Guide

1. Install Node.js

- Download from <https://nodejs.org/>
- Verify with:
node -v
npm -v

2. Install Git

- Download from <https://git-scm.com/>
- Verify: git --version

3. Install VS Code

- Download from <https://code.visualstudio.com/>

4. Install Expo CLI

npm install -g expo-cli
Check with: expo --version

5. Create a New React Native Project

expo init MyLoginApp
Choose template: *blank (JavaScript)*
cd MyLoginApp
npm start

6. Install Expo Go on Phone

- Download from App Store (iOS) or Google Play (Android).
- Scan the QR code shown in the terminal/browser when running npm start.

2. Example Login App Code

```
import React, { useState } from 'react';
import { View, Text, TextInput, Button, StyleSheet } from 'react-native';

export default function App() {
  const [username, setUsername] = useState('');
  const [password, setPassword] = useState('');

  const handleLogin = () => {
    alert(`Username: ${username}, Password: ${password}`);
  };

  return (
    <View style={styles.container}>
      <Text style={styles.title}>Login</Text>
      <TextInput
        style={styles.input}
        placeholder="Enter username"
        value={username}
        onChangeText={setUsername}
      />
      <TextInput
        style={styles.input}
        placeholder="Enter password"
        value={password}
        onChangeText={setPassword}
        secureTextEntry
      />
      <Button title="Login" onPress={handleLogin} />
    </View>
  );
}
```

```

    });
  }

  const styles = StyleSheet.create({
    container: {
      flex: 1,
      justifyContent: 'center',
      padding: 20,
      backgroundColor: '#f2f2f2',
    },
    title: {
      fontSize: 28,
      marginBottom: 20,
      textAlign: 'center',
      fontWeight: 'bold',
    },
    input: {
      height: 50,
      borderWidth: 1,
      borderColor: '#ccc',
      borderRadius: 8,
      paddingHorizontal: 10,
      marginBottom: 15,
      backgroundColor: 'fff',
    },
  });
});

```

React Functions & Components Used:

- **useState:** A React Hook that allows us to create state variables inside functional components. Example: `const [username, setUsername] = useState("")` stores the entered username.
- **View:** A container component, similar to a `<div>` in HTML. Used for layout.
- **Text:** Displays text, similar to `<p>` or `<span>` in HTML.
- **TextInput:** Input field for user text entry. We bind it with state using `value` and `onChangeText`.
- **Button:** A clickable button. `onPress` defines what happens when pressed.
- **StyleSheet:** Used to define CSS-like styles in React Native. Each style is written as a JavaScript object.
- **alert():** Simple popup message to display entered username and password.

Running the App on Phone:

1. Run `npm start` in project folder.
2. A QR code will open in your browser.
3. Open the **Expo Go** app on your phone.
4. Scan the QR code → App runs instantly on your phone.

Next Steps:

- You can edit `App.js` in VS Code and see live updates on your phone.
- Later, you can build an APK/IPA to install the app outside Expo.

5. Understanding npm, npx, and Expo CLI:

npm (Node Package Manager):

- Installs, updates, and manages packages for Node.js projects.
- Stores them in `node_modules` and tracks them in `package.json`.

- Example: npm install react-native, npm install -g expo-cli.

expo-cli (Expo Command Line Interface):

- A developer tool for creating, running, and building React Native apps.
- Works with the Expo Go app to preview apps on your phone.
- Example: expo start, expo build.
- Global installation is now deprecated; use npx instead.

npx (Node Package eXecute):

- Runs npm packages without installing them globally.
- Always uses the latest version of the package.
- Example: npx create-expo-app MyApp, npx expo start.

Summary Table:

Tool	Full Name	Use	npm	Node Package Manager	Installs/manages libraries & tools	expo-cli	Expo
Expo	Expo	CLI tool for creating & running React Native projects (deprecated globally)					
npx	Node Package eXecute	Runs npm packages without global install (recommended way)					